

Lightweight Cryptography for the Internet of Things: A Multi-Metric Experimental Assessment

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Abstract—The increasing use of resource-constrained devices in IoT applications demands cryptographic algorithms that are both secure and efficient. This work presents a thorough comparative analysis of the most widely cited lightweight symmetric cryptographic algorithms—ASCN, XTEA, SLIM, AES-LWC, SHADOW, and PRESENT—using the original AES as a benchmark due to its proven security and adaptability in both hardware and software. The evaluation was conducted on ESP32 and Raspberry Pi Pico platforms, considering execution time, throughput, memory usage, energy consumption, avalanche effect, bit balance, and NIST statistical tests. XTEA achieved the best overall performance, combining high efficiency with full compliance with all security criteria. SLIM and AES-LWC demonstrated balanced performance across both efficiency and security. ASCN was efficient but exhibited the highest memory usage among all algorithms. SHADOW showed intermediate results, while PRESENT performed the worst, with low efficiency and failures in statistical tests. These findings provide practical guidance for selecting lightweight ciphers that meet the specific performance and security constraints of IoT applications.

Index Terms—internet of things, lightweight cryptography, experimental evaluation, energy consumption, security metrics

I. INTRODUCTION

The Internet of Things (IoT) refers to physical devices that are embedded with sensors, software and network connectivity, allowing them to collect and share data. IoT devices, also known as smart objects, can range from simple objects, like smart thermostats and wearables, to complex industrial machinery and smart cities [1]. By 2030, the number of IoT applications is projected to reach 40 billion devices, representing a substantial increase from previous years. [2].

As the adoption of IoT devices expands, concerns about security and privacy have grown. Many IoT devices are vulnerable to cyberattacks, which may compromise the confidentiality and integrity of sensitive information. Moreover, their extensive data collection capabilities raise significant issues related to privacy and data protection [1].

Secure communication over the Internet is a persistent challenge that demands robust and dependable solutions. Cryptography directly addresses this need, offering the most effective approach for protecting sensitive information during

transmission. However, cryptographic algorithms are computationally intensive, impacting key performance factors such as execution time, memory footprint, and energy savings, which is a major challenge for IoT applications, as most of them are deployed on resource-constrained devices and powered by batteries [3].

The growing use of IoT devices has led to the development of lightweight cryptographic algorithms, which aim to reduce memory usage and computational complexity without compromising security. These algorithms are specifically designed to meet the constraints of resource-limited embedded environments, offering secure and efficient encryption for IoT applications [4], [5].

Some works, such as [6], [7], [8], [9], and [10], present and compare various lightweight cryptographic algorithms in terms of memory footprint, execution time, and throughput across several platforms commonly used in IoT applications. However, it is important to highlight that these studies do not include an experimental security analysis of the evaluated algorithms, such as avalanche effect, balance, or the NIST tests. A more detailed discussion of each will be provided in the Related Work section.

The objective of this work is to provide a comparative analysis of the symmetric lightweight cryptographic algorithms ASCN, XTEA, SLIM, AES-LWC (as proposed in [11]), SHADOW, and PRESENT, together with the original AES, which serves as a reference due to its well-established status. These algorithms were selected for being among the most cited in recent lightweight cryptography studies, along with AES and its lightweight variant, AES-LWC. The evaluation considers memory footprint, execution time, energy consumption, and throughput on the ESP32 and Raspberry Pi Pico platforms. Unlike previous studies, this work also includes an experimental security assessment based on the avalanche effect, bit balance, and NIST statistical tests. This approach addresses important gaps in the literature and, for the first time, comparatively evaluates SHADOW, SLIM, and AES-LWC.

The structure of this paper is as follows: Section II reviews related work; Section III presents the methodology; Section IV describes the selected algorithms; Section V discusses the results; and Section VI concludes the paper.

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II. RELATED WORK

This section presents and discusses related works that evaluate lightweight cryptographic algorithms in the context of IoT applications.

In [6], the authors conduct an experimental performance analysis of various cryptographic algorithms in the context of IoT applications: symmetric block ciphers (DES, PRESENT, PICCOLO, AES, CLEFIA, IDEA), stream ciphers (SALSA20, XCHACHA), hashing techniques (SHA, MD5), and asymmetric cryptography (RSA). The main evaluation metrics considered are memory footprint and execution time. The tests were carried out on three platforms: Raspberry Pi 3B, Raspberry Pi Zero W, and Arduino MKR1300. The results show that algorithm performance varies notably across platforms, with DES and XCHACHA achieving the best throughput on Raspberry Pi, while DES and MD5 performed best on Arduino.

The authors in [7] evaluate the performance of the lightweight cryptographic algorithms PRESENT, SIMON, and XTEA in a real-world IoT environment, with a focus on image encryption. The algorithms were implemented on a Raspberry Pi 3, capturing images via webcam and transmitting the encrypted data over a wireless intranet. The evaluation considered the execution time of the algorithms. SIMON demonstrated the best performance, with significantly lower encryption time compared to PRESENT and XTEA, while all algorithms preserved image quality with minimal distortion. The authors conclude that SIMON is the most efficient choice for latency-sensitive IoT applications involving image transmission.

In [8], the authors analyzed the performance of the lightweight cryptographic algorithms ASCON and Grain128-AEAD in a simulated IoT environment based on the MQTT protocol. The encryption time was used as a metric. Results showed that encryption significantly increased processing time using ASCON, but the added latency was considered acceptable given the enhanced data confidentiality and integrity. ASCON slightly outperformed Grain128-AEAD in all scenarios. The authors conclude that despite the performance overhead, these lightweight algorithms are suitable for securing MQTT-based IoT systems with minimal impact on efficiency.

The authors in [9] present a comprehensive evaluation of eleven symmetric cryptographic algorithms on ARM Cortex-M microcontrollers, targeting IoT applications. The tested algorithms include AES, IDEA, ARIA, NOEKEON, CAMELLIA, SEED, GOST, SIMON, HIGHT, SPECK, and PRESENT. Experiments were conducted on Cortex-M3, M4, M7, and M33 platforms. The evaluation metrics include execution time, power consumption, and memory footprint. The results indicate that GOST has the smallest memory footprint, while SEED is the most energy-efficient and demonstrates superior performance across all ARM Cortex-M boards.

In [10], the authors present a performance evaluation, in terms of throughput, of the lightweight cryptographic algorithms ASCON, TinyJambu, and Xoodoo, implemented on the Raspberry Pi Pico W to secure IoT data transmission

in energy-efficient applications. ASCON showed the lower throughput, while TinyJambu achieved the best performance with a compact code size ideal for resource-constrained devices. Xoodoo provided moderate performance, but with a larger memory footprint. All algorithms enabled secure end-to-end communication, demonstrating their suitability for IoT systems where energy efficiency is critical. Table I summarizes the related works.

TABLE I
SUMMARY OF RELATED WORK EVALUATING LIGHTWEIGHT CRYPTOGRAPHIC ALGORITHMS AND METRICS FOR IoT APPLICATIONS.

Ref.	Algorithms	Memory Footprint	Exec. Time	Energy Cons.	Throughput	Security	Platforms
6	AES, CLEFIA, DES, IDEA, MD5, PICCOLO, PRESENT, RSA, SALSA20, SHA, XCHACHA20	✓	✓	✗	✗	✗	RASP. PI 3 RASP. PI ZERO W ARDUINO MKR1300
7	PRESENT, SIMON, XTEA	✗	✓	✗	✗	✗	RASP. PI 3
8	ASCON, GRAIN-128	✗	✓	✗	✗	✗	SIMULATION
9	AES, ARIA, CAMELLIA, GOST, HIGHT, IDEA, NOEKEON, PRESENT, SEED, SIMON, SPECK	✓	✓	POWER CONSUMPT. (W)	✗	✗	ARM M3 ARM M4 ARM M7 ARM M33
10	ASCON, TINYJAMBU, XOODYAK	✗	✗	✗	✓	✗	RASP. PI PICO W

III. METHODOLOGY

This section presents the systematic literature review, the experimental platforms, and the procedures for measuring energy consumption.

A. Systematic Literature Review Approach

A systematic literature review was conducted across major scientific databases to identify the most commonly adopted lightweight cryptographic algorithms in IoT applications. The goal was to determine which algorithms have received the most attention in recent academic research. The search query used was: ((“Internet of Things” OR “IoT”) AND (“Lightweight Cryptography” OR “Lightweight Security”)).

To ensure the relevance and quality of the selected studies, a four-stage filtering process was applied. The first filter limited the results to papers published between 2021 and 2025. The second filter removed duplicate entries across the three databases (ACM, IEEE, and Scopus). The third filter applied specific exclusion criteria, eliminating papers that focused solely on cryptographic attacks, were outside the IoT context, dealt only with key generation, did not address cryptographic algorithms directly, focused exclusively on authentication, used stream ciphers instead of block ciphers, were literature surveys, or were unavailable for access.

The algorithms ANU-II, CRAFT, ISL-LWS, KLEIN, LEA, LOONG, MCRYPTON, PRIDE, PRINCE, RC5, SHC, SIMON, SIMECK, SKINNY, TINYJAMBU, and XOODYAK appeared only once across the selected papers. Therefore, these algorithms and their corresponding papers were excluded, and only those that mentioned algorithms appearing in at least two different studies were retained. The selected algorithms are:

AES, ASCON, PRESENT, SHADOW, SLIM, and TEA. This step constituted the fourth and final filtering stage. Table II provides a summary of the number of papers remaining at each stage of the filtering process. After applying the fourth filter, 23 papers remained — 17 from IEEE and 6 from Scopus — representing almost 60% of the total number of papers remaining after the third filtering stage. Among these, PRESENT and AES each appeared in 6 papers (15.38%), ASCON in 4 (10.26%), TEA in 3 (7.69%), and SHADOW and SLIM in 2 papers each (5.13%).

TABLE II
NUMBER OF PAPERS REMAINING AFTER EACH FILTERING STAGE OF THE
SYSTEMATIC LITERATURE REVIEW.

	Initial Result	1st Filter	2nd Filter	3rd Filter	4th Filter
ACM	232	133	132	0	0
IEEE	507	285	285	24	17
SCOPUS	1003	593	358	15	6
TOTAL	1742	1011	775	39	23

Among the algorithms selected through the literature review process, some had already been addressed in related works, while others had not. The algorithms that had not yet appeared in any comparative study are AES-LWC, SHADOW, and SLIM. However, even for those previously analyzed in related works, there were still significant gaps in the evaluation, especially regarding security aspects. For example, AES and PRESENT lacked analysis regarding energy consumption, throughput, and experimental security evaluation. ASCON required evaluation of memory usage, energy consumption, and security metrics. XTEA lacked analysis in three aspects: memory footprint, throughput, energy consumption, and had not been subjected to any experimental security assessment. Since they had not been previously compared, AES-LWC, SHADOW, and SLIM underwent a full evaluation process, including execution time, memory footprint, energy consumption, throughput, as well as security analyses such as avalanche effect, bit balance, and NIST randomness tests. Table III summarizes all the contributions presented in this paper.

TABLE III
SUMMARY OF THE NEW CONTRIBUTIONS INTRODUCED BY THIS PAPER.

Algorithm	Execution Time	Memory Footprint	Energy Consumption	Throughput	Security
AES	-	-	✓	✓	✓
AES-LWC	✓	✓	✓	✓	✓
ASCON	-	✓	✓	-	✓
PRESENT	-	-	✓	✓	✓
SHADOW	✓	✓	✓	✓	✓
SLIM	✓	✓	✓	✓	✓
XTEA	-	✓	✓	✓	✓

B. Embedded Platforms Used

The experiments were carried out on two platforms commonly used in the development of IoT applications: the Raspberry Pi Pico by Raspberry Pi and the ESP-WROOM-32 Devkit V1 by Espressif.

The Raspberry Pi Pico is a widely adopted platform in IoT projects, known for its cost-effective processor [18]. It is based on the RP2040 microcontroller, developed by Raspberry Pi, which integrates an Arm Cortex-M0+ processor operating at up to 133 MHz. The board includes 264 KB of RAM and 2 MB of flash memory.

The ESP-WROOM-32 is another popular and affordable platform for IoT development [19]. It features a 32-bit dual-core Xtensa processor, 520 KB of RAM, and 4 MB of flash memory, and can reach a maximum clock frequency of 240 MHz.

C. Power and Energy Evaluation

To carry out the energy consumption analysis, the Power Profiler Kit II (PPK2) from Nordic Semiconductor was employed. It is a high resolution instrumentation tool specifically designed for monitoring current in embedded systems [20]. Thanks to its 100nA resolution and wide dynamic range from 200 nA to 1 A, the PPK2 enables accurate profiling of power usage. When operating in source mode, it supplies voltages configurable between 0.8V and 5V. With a sampling rate of 100 KS/s, the device captures a current sample every 10 μ s, making it suitable for detailed temporal analysis of energy consumption.

In the experimental setup, both platforms were connected in series with the PPK2 to enable current sampling. The output voltage was configured to 3.3V, and the sampling rate was set to 100,000 samples per second. Subsequently, each cryptographic algorithm was executed 100 times sequentially on each platform, and the corresponding CSV files containing the captured current samples were generated.

To calculate the total energy consumption, each current sample from the CSV file was multiplied by the supply voltage (3.3V) to obtain the instantaneous power. These values were then multiplied by the sampling interval (10 μ s) to compute instantaneous energy. The total energy was obtained by summing all resulting values.

IV. LIGHTWEIGHT CRYPTOGRAPHIC ALGORITHMS

This section provides an overview of each selected cryptographic algorithm, highlighting their main characteristics, such as structure, key and block sizes, and the number of rounds.

The Advanced Encryption Standard (AES) is a symmetric block cipher standardized by NIST in 2001 as the successor to the Data Encryption Standard (DES). In this work, the AES-128 variant is adopted, operating on 128-bit blocks with 128-bit keys across 10 rounds, each consisting of SubBytes, ShiftRows, MixColumns, and AddRoundKey. AES is widely adopted due to its proven security, efficiency in both hardware and software implementations, and broad applicability in commercial and governmental systems [12]. AES-LWC is a lightweight variant of AES that optimizes the SubBytes stage by using a smaller s-box, and the MixColumns stage by employing simpler and less computationally intensive operations, while preserving the same block size (128 bits), key size (128 bits), and number of rounds (10) as AES-128 [11].

ASCON is a family of lightweight authenticated encryption algorithms designed for high efficiency in constrained environments, such as IoT devices. In this work, ASCON is specified with a 128-bit block, a 128-bit key, and 12 rounds. It supports both encryption and message authentication through a sponge-based construction and a permutation-based design. Its balance between security, performance, and low resource usage makes it well-suited for modern embedded applications [13].

PRESENT is a lightweight block cipher introduced in 2007, specifically designed for environments with severe resource constraints such as RFID devices. It operates on 64-bit blocks with 128-bit keys and applies 31 rounds of simple operations, including a substitution layer using a 4-bit s-box and a bitwise permutation layer. These features make it highly compact and hardware-efficient while still offering reasonable security for lightweight applications [14].

SHADOW is a lightweight block cipher proposed in 2007, designed to offer strong cryptographic security with minimal hardware complexity. In this work, it is specified with 64-bit blocks, 128-bit keys, and 32 rounds. SHADOW combines modular addition, s-box substitution, and bit-level permutation in a generalized Feistel structure, optimized for low-area and low-power implementations [15].

SLIM is a lightweight block cipher designed specifically for the Internet of Health Things (IoHT), addressing the need for secure and efficient communication in low-resource medical devices. In this work, SLIM operates on 128-bit blocks with 128-bit keys and 32 rounds, implemented as a substitution-permutation network with 4-bit s-boxes and a linear diffusion layer. Its design emphasizes low power consumption and small footprint, making it suitable for wearable and implantable healthcare systems [16].

XTEA (eXtended Tiny Encryption Algorithm) is a lightweight block cipher introduced in 1997 as an improvement over the original TEA. It operates on 64-bit blocks with a 128-bit key using a Feistel structure with modular additions, XORs, and shifts. Although the original specification recommends 64 rounds, in this work the 32-round variant is specified. XTEA maintains simplicity and compactness, making it suitable for embedded systems and resource-constrained devices [17]. Table IV summarizes the main parameters of the evaluated algorithms.

TABLE IV
BLOCK SIZE, KEY SIZE, AND NUMBER OF ROUNDS OF THE EVALUATED
LIGHTWEIGHT CRYPTOGRAPHIC ALGORITHMS.

Algorithm	Block (bits)	Key (bits)	Rounds
AES	128	128	10
AES-LWC	128	128	10
ASCON	128	128	12
PRESENT	64	128	31
SHADOW	64	128	32
SLIM	128	128	32
XTEA	64	128	32

V. RESULTS

This section presents the results for execution time, throughput, memory footprint, energy consumption, and security analyses including avalanche effect, bit balance, and NIST statistical tests for AES and the selected lightweight algorithms on the ESP32 and Raspberry Pi Pico platforms.

A. Execution Time and Throughput Analysis

The execution time of each algorithm was initially measured on both platforms. To obtain these values, the `micros()` function from the Arduino IDE was used to record the elapsed time during the encryption process. Each algorithm was executed 100 times in sequence, and the average execution time was calculated using the arithmetic mean. The input used in all tests was a randomly generated string with the exact size of one block (16 bytes). Table V summarizes the results obtained for each algorithm across both platforms.

TABLE V
AVERAGE EXECUTION TIME (IN MICROSECONDS) OF EACH ALGORITHM
ON ESP32 AND RASPBERRY PI PICO.

Algorithm	ESP32 (μs)	Raspberry Pi Pico (μs)
XTEA	32	28
ASCON	124	181
SLIM	157	152
AES-LWC	411	422
SHADOW	892	866
AES	1519	1802
PRESENT	2360	2306

As shown in Table V, some algorithms exhibited lower execution times on the ESP32 (AES, AES-LWC, and ASCON), while others performed better on the Raspberry Pi Pico (PRESENT, SHADOW, SLIM, and XTEA). Although in most cases the execution time differences between the platforms were small, some algorithms showed more noticeable discrepancies, for instance, AES, which presented a difference of up to 283μs.

It is worth noting in Table V that, despite being designed as a lightweight cryptographic algorithm, PRESENT exhibits a longer execution time than AES, which was not developed with this purpose in mind. This is mainly due to its hardware-oriented design, which relies on bit-level permutations that are efficient in hardware but computationally expensive in software. Moreover, PRESENT performs 31 rounds, further increasing the processing time in software implementations. A longer execution time may lead to higher energy consumption, which is undesirable for IoT applications.

In terms of throughput, the value was computed based on Equation 1:

$$\text{Throughput (kbps)} = \left(\frac{128}{\text{Exec. Time (μs)}} \right) \times 1000 \quad (1)$$

Table VI summarizes the throughput results for each cryptographic algorithm across both platforms.

TABLE VI
CALCULATED THROUGHPUT (IN KBPS) OF EACH ALGORITHM.

Algorithm	ESP32 (kbps)	Raspberry Pi Pico (kbps)
XTEA	4000.00	4571.43
ASCON	1032.26	707.18
SLIM	815.29	842.11
AES-LWC	311.44	303.32
SHADOW	143.50	147.81
AES	84.27	71.03
PRESENT	54.24	55.51

As shown in Table VI, the obtained values are directly related to the execution time. Since all tests used the same input size (128 bits), lower execution times result in higher throughput.

B. Memory Footprint Analysis

The memory footprint for each cryptographic algorithm was assessed using the Arduino IDE, which, upon sketch verification, reports both program and dynamic memory usage. Dynamic memory (data) holds variables and pointers used during program execution, whereas program memory stores the application's source code [21].

The chart shown in Figure 1 presents the memory consumption of each algorithm, including both program memory and dynamic memory, on both platforms.

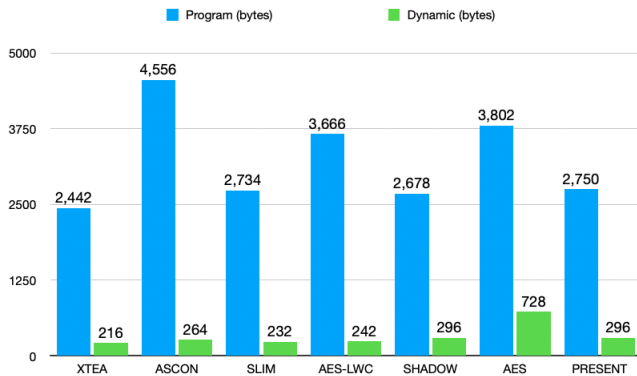


Fig. 1. Program and dynamic memory usage of each cryptographic algorithm.

As shown in Figure 1, the XTEA algorithm consumes the least amount of memory, both program and dynamic. On the other hand, ASCON, although designed as a lightweight cryptographic algorithm, presents the highest program memory usage, even surpassing AES. This is mainly due to its sponge-based construction with a 320-bit internal state and complex permutation functions, which increase code size in software implementations. Such demand may be unsuitable for IoT applications with strict memory constraints.

C. Energy Consumption Analysis

The total energy dissipation values are presented in the chart in Figure 2, calculated from the electric current samples collected using the PPK2.

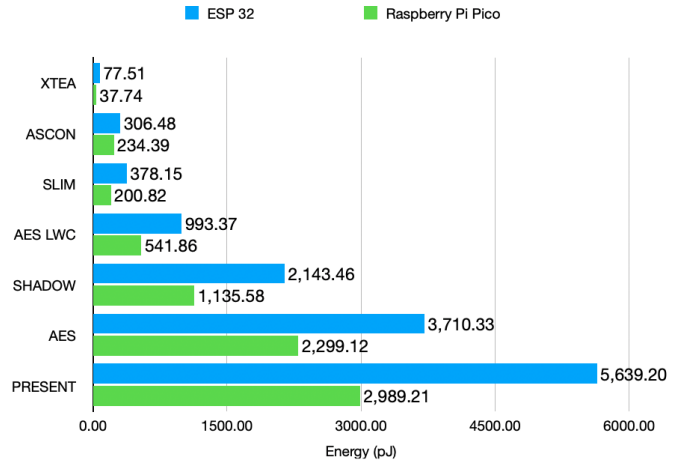


Fig. 2. Total energy consumption (in picojoules) for each algorithm.

It is interesting to observe in the chart in Figure 2 that the total energy dissipated is strongly correlated with execution time. However, when comparing with the Table V, we can see that the AES, AES-LWC, and ASCON algorithms achieved lower execution times on the ESP32, while PRESENT, SHADOW, SLIM, and XTEA performed better on the Raspberry Pi Pico. Nevertheless, in terms of energy consumption, all algorithms exhibited lower energy usage on the Raspberry Pi Pico. This behavior can be explained by architectural differences between the platforms. The ESP32, although faster, has a higher baseline power consumption due to its more complex dual-core design and integrated peripherals. In contrast, the Raspberry Pi Pico uses an energy-efficient Cortex-M0+ processor, which results in lower overall energy usage even when execution times are slightly longer.

It is also worth noting that the algorithms exhibited different variations in energy consumption between the platforms. For example, the ASCON algorithm consumed 23.52% more energy on the ESP32, while the XTEA algorithm showed an increase of 51.31%.

D. Security Analysis

Three different tests were employed to evaluate the security properties of the ciphertexts generated by each algorithm: the avalanche effect, the bit balance criterion, and the NIST statistical tests.

The avalanche effect assesses how a single-bit change in the input affects the ciphertext, with ideal algorithms altering about half the output bits. The bit balance criterion checks whether the ciphertext contains roughly equal numbers of zeros and ones, ensuring the absence of statistical biases that could weaken security. Figure 3 shows the results for avalanche effect and bit balance obtained for each algorithm.

The algorithms presented a very similar distribution of zeros and ones in their ciphertexts. However, in terms of the avalanche effect, ASCON and SLIM exhibited values significantly below 50%, with observed averages of approximately 12% and 20%, respectively.

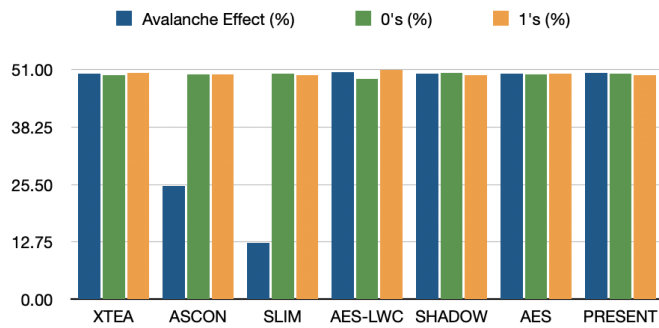


Fig. 3. Avalanche effect percentage and distribution of zeros and ones in the ciphertext (bit balance) for each algorithm.

The NIST statistical test suite, consisting of fifteen tests, was also applied to the ciphertexts to evaluate their randomness and detect possible statistical weaknesses. Among the selected algorithms, only PRESENT failed multiple tests from the NIST Statistical Test Suite, indicating non-random patterns in its ciphertexts. The other algorithms passed all applied tests, confirming adequate statistical behavior.

VI. CONCLUSIONS

This work presented a comprehensive experimental evaluation of seven symmetric cryptographic algorithms: AES, AES-LWC, ASCON, PRESENT, SHADOW, SLIM, and XTEA, executed on two representative IoT platforms, ESP32 and Raspberry Pi Pico. The evaluation considered execution time, throughput, memory usage, and energy consumption, along with a practical security analysis based on avalanche effect, bit balance, and NIST statistical tests.

Among all evaluated algorithms, XTEA was clearly the most efficient. It achieved the best results in execution time, throughput, memory usage, and energy consumption, while also passing all security tests. These results make XTEA the most suitable option for highly constrained IoT devices.

SLIM and AES-LWC also demonstrated solid and well-balanced performance. SLIM consumed little energy, had good speed, and low memory usage, although its avalanche effect was below the ideal threshold. AES-LWC stood out for its excellent security profile and balanced performance across all metrics, despite not being the fastest.

ASCON, despite being the NIST lightweight cryptography standard, showed high memory consumption, which can be a limitation for very restricted environments. It maintained acceptable performance in speed and security, but is not the best choice when memory is a critical constraint.

SHADOW presented intermediate results, with moderate energy and memory usage and good security, but it was clearly outperformed by XTEA and SLIM in most metrics.

Finally, PRESENT showed the worst performance overall. It had the slowest execution time, the lowest throughput, and was the only algorithm that failed the NIST statistical tests. Surprisingly, it performed worse than the standard AES in almost all aspects, indicating it may no longer be a viable option for modern IoT applications.

These results reinforce the importance of evaluating cryptographic algorithms using a multi-metric approach. While security remains essential, performance and energy efficiency are equally critical in embedded scenarios. As a future direction, the inclusion of side-channel analysis is proposed to assess the resistance of each cipher to attacks in realistic scenarios.

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